

Assignment 4

Objective: Practice defining classes, friend functions, overloading operators

Assignment Requirements

1. Define BankAccount Class:

- Private member variables:
 - `accountNumber`: A unique account number for each account.
 - `balance`: The current balance in the account.
- Public member functions:
 - **Constructors**: To initialize accounts with an account number and optional initial balance.
 - **Default Constructor**: define default values.
 - **Getters and Setters**: Get the account number and balance and set the account number and balance.
 - **Deposit**: To deposit an amount into the account.
 - **Withdraw**: To withdraw an amount from the account (if sufficient balance is available).
- **Friend Functions**:
 - A friend function to transfer an amount from one account to another.
- **Overloaded Operators**:
 - `+`: To add an amount to the account balance.
 - `-`: To withdraw an amount (if sufficient balance).
 - `==` and `!=`: To compare two accounts by their balances.
 - `<<` and `>>`: For outputting and inputting account details.

2. Friend Function for Balance Transfer:

- A friend function should be defined to transfer funds between two `BankAccount` objects.
- A friend function should access private data members directly.
- The function should check that the source account has sufficient balance before transferring.

3. Main Function for Testing:

- Create instances of `BankAccount1` and `BankAccount2` to test all functionalities:
 - Perform deposits, withdrawals, balance comparisons, and balance transfers.
- Use the overloaded operators and friend functions to demonstrate functionality.

Submission:

Submit the C++ source file (.cpp) and the screenshot of your program output on Moodle.